

The Fundamental Theorem of Calculus

ANSWER KEY



FTC Part 2: evaluating definite integrals

- 1 Evaluate $\int_0^3 (x^2 - 2x) dx$ using the Fundamental Theorem of Calculus.

$$F(x) = \frac{x^3}{3} - x^2. F(3) - F(0) = (9 - 9) - (0) = 0.$$

- 2 Evaluate $\int_1^4 \sqrt{x} dx$.

$$F(x) = \frac{2}{3}x^{3/2}. F(4) - F(1) = \frac{2}{3}(8) - \frac{2}{3}(1) = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}.$$

- 3 Evaluate $\int_0^{\pi/2} \cos x dx$.

$$F(x) = \sin x. F(\pi/2) - F(0) = 1 - 0 = 1.$$

FTC Part 1: accumulation functions

- 4 Let $g(x) = \int_0^x t^2 dt$. Find $g'(x)$ using FTC Part 1.

By FTC Part 1, $g'(x) = x^2$ (the integrand evaluated at x).

- 5 Let $g(x) = \int_1^x \sin(t^2) dt$. Find $g'(2)$.

$$g'(x) = \sin(x^2), \text{ so } g'(2) = \sin(4) \approx -0.757.$$

- 6 Let $g(x) = \int_0^{x^2} e^{-t} dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.

$$g'(x) = e^{-x^2} \cdot \frac{d}{dx}(x^2) = 2xe^{-x^2}.$$

Interpreting accumulation functions

- 7 A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau) d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.

$$s(4) = \int_0^4 (3 - 2\tau) d\tau = [3\tau - \tau^2]_0^4 = (12 - 16) - 0 = -4. \text{ This means the particle's net displacement from } t = 0 \text{ to } t = 4 \text{ is } -4 \text{ m (it ends up 4 m to the left/behind its starting point).}$$

- 8 Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t) dt$, explain what feature of g 's graph occurs where f crosses the x -axis, and where f has a local max.

Where f crosses the x -axis (changes sign), $g' = f$ changes sign, so g has a local extremum there. Where f has a local max, $g' = f$ has a local max, meaning g has an inflection point there (since $g'' = f'$ changes sign from $+$ to $-$).

Average value of a function

- 9 Find the average value of $f(x) = x^2$ on $[0, 3]$.

$$\text{Average} = \frac{1}{3-0} \int_0^3 x^2 dx = \frac{1}{3} \left[\frac{x^3}{3} \right]_0^3 = \frac{1}{3}(9) = 3.$$

- 10 The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$. Find the average temperature over this period.

$$\text{Average} = \frac{1}{24} \int_0^{24} [70 + 10\sin(\frac{\pi t}{12})] dt. \text{ The sine term integrates to 0 over a full period (24 hours = 2 full cycles), so the average} = 70^\circ.$$

Definite integrals with substitution-friendly integrands

- 11 Evaluate $\int_0^1 2x(x^2 + 1)^3 dx$ using $u = x^2 + 1$.

$$du = 2x dx. \text{ When } x = 0, u = 1; \text{ when } x = 1, u = 2. \int_1^2 u^3 du = \left[\frac{u^4}{4} \right]_1^2 = \frac{16}{4} - \frac{1}{4} = \frac{15}{4}.$$

The Mean Value Theorem for integrals

- 12 Find the value c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(c)$ equals the average value found earlier.

The average value of $f(x) = x^2$ on $[0, 3]$ is 3 (found earlier). Set $f(c) = c^2 = 3$: $c = \sqrt{3} \approx 1.73$, which lies in $(0, 3)$.

Visualizing FTC Part 1 as an accumulation function

- 13 The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2 dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x .

Since $f(t) = t^2 \geq 0$ for all t , every strip added as x increases contributes non-negative area, so $g(x) = \int_0^x t^2 dt$ is always increasing (or constant only at $x = 0$). By FTC Part 1, $g'(x)$ equals the height of the integrand curve at x , i.e. $g'(x) = f(x) = x^2$, matching the graph's height directly.

A capstone FTC application

- 14 Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \leq t \leq 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.

Volume $= \int_0^6 (8 - t) dt = \left[8t - \frac{t^2}{2} \right]_0^6 = 48 - 18 = 30$ liters. Since $r(t) = 8 - t = 0$ at $t = 8$, the inflow rate drops to zero exactly when the tank presumably reaches capacity, consistent with the water level leveling off.

FTC Part 1 with more complex bounds

- 15 Let $g(x) = \int_3^{2x} \ln(t + 1) dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.

$$g'(x) = \ln(2x + 1) \cdot \frac{d}{dx}(2x) = 2\ln(2x + 1).$$

A second average-value application

- 16 A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \leq t \leq 8$. Find the average production rate over the shift.

Average

$$= \frac{1}{8} \int_0^8 (50 + 10t - t^2) dt = \frac{1}{8} \left[50t + 5t^2 - \frac{t^3}{3} \right]_0^8 = \frac{1}{8} \left(400 + 320 - \frac{512}{3} \right) = \frac{1}{8} \left(\frac{2160 - 512}{3} \right) = \frac{1648}{24} \approx 68.67$$

units/hour.

Multiple choice — easy 15

- 17 Evaluate $\int_2^6 (6x^2 + 8x) dx$ using the Fundamental Theorem of Calculus.

a. 544 ✓

b. 546

c. 576

d. 542

$$F(x) = 2x^3 + 4x^2. F(6) - F(2) = (576) - 32 = 544.$$

18 Evaluate $\int_1^5 (6x^2 + 6x)dx$ using the Fundamental Theorem of Calculus.

a. 320 ✓

b. 318

c. 322

d. 325

$$F(x) = 2x^3 + 3x^2. F(5) - F(1) = (325) - 5 = 320.$$

19 Evaluate $\int_{-1}^0 (3x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

a. -2

b. 2

c. 3

d. 0 ✓

$$F(x) = x^3 + x^2. F(0) - F(-1) = (0) = 0.$$

20 Evaluate $\int_2^4 (9x^2)dx$ using the Fundamental Theorem of Calculus.

a. 170

b. 168 ✓

c. 166

d. 192

$$F(x) = 3x^3. F(4) - F(2) = (192) - 24 = 168.$$

21 Evaluate $\int_0^2 (-6x^2 - 2x)dx$ using the Fundamental Theorem of Calculus.

a. -18

b. -22

c. -17

d. -20 ✓

$$F(x) = -2x^3 - x^2. F(2) - F(0) = (-20) = -20.$$

22 Evaluate $\int_2^6 (6x^2 - 6x)dx$ using the Fundamental Theorem of Calculus.

a. 318

b. 320 ✓

c. 322

d. 324

$$F(x) = 2x^3 - 3x^2. F(6) - F(2) = (324) - 4 = 320.$$

23 Evaluate $\int_{-2}^1 (3x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

a. 4

b. 8

c. 2

d. 6 ✓

$$F(x) = x^3 + x^2. F(1) - F(-2) = (2) + 4 = 6.$$

24 Evaluate $\int_2^7 (-3x^2 - 8x)dx$ using the Fundamental Theorem of Calculus.

a. -513

b. -539

c. -517

d. -515 ✓

$$F(x) = -x^3 - 4x^2. F(7) - F(2) = (-539) + 24 = -515.$$

25 Evaluate $\int_2^7 (6x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

a. 715 ✓

b. 735

c. 717

d. 713

$$F(x) = 2x^3 + x^2. F(7) - F(2) = (735) - 20 = 715.$$

26 Evaluate $\int_0^3 (6x^2 - 2x)dx$ using the Fundamental Theorem of Calculus.

a. 45 ✓

b. 43

c. 48

d. 47

$$F(x) = 2x^3 - x^2. F(3) - F(0) = (45) = 45.$$

27 Evaluate $\int_{-2}^3 (6x^2)dx$ using the Fundamental Theorem of Calculus.

a. 70 ✓

b. 72

c. 54

d. 68

$$F(x) = 2x^3. F(3) - F(-2) = (54) + 16 = 70.$$

28 Evaluate $\int_{-2}^2 (-3x^2 - 2x)dx$ using the Fundamental Theorem of Calculus.

a. -12

b. -16 ✓

c. -14

d. -18

$$F(x) = -x^3 - x^2. F(2) - F(-2) = (-12) - 4 = -16.$$

29 Evaluate $\int_0^5 (6x^2 + 4x)dx$ using the Fundamental Theorem of Calculus.

a. 298

b. 303

c. 302

d. 300 ✓

$$F(x) = 2x^3 + 2x^2. F(5) - F(0) = (300) = 300.$$

30 Evaluate $\int_1^4 (9x^2 + 6x)dx$ using the Fundamental Theorem of Calculus.

a. 232

b. 234 ✓

c. 240

d. 236

$$F(x) = 3x^3 + 3x^2. F(4) - F(1) = (240) - 6 = 234.$$

31 Evaluate $\int_2^5 (3x^2 - 6x)dx$ using the Fundamental Theorem of Calculus.

a. 50

b. 54 ✓

c. 56

d. 52

$$F(x) = x^3 - 3x^2. F(5) - F(2) = (50) + 4 = 54.$$

Multiple choice — medium 15

32 Let $g(x) = \int_{-1}^{-4x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

- a. 144
- b. -36
- c. -580
- d. -576 ✓

$$g'(x) = (-4x)^2 \cdot (-4) = (16x^2)(-4), \text{ so } g'(3) = (-12)^2 \cdot (-4) = 144(-4) = -576.$$

33 Let $g(x) = \int_{-3}^{-4x} t^2 dt$. Find $g'(2)$ using FTC Part 1 and the Chain Rule.

- a. -16
- b. -256 ✓
- c. 64
- d. -260

$$g'(x) = (-4x)^2 \cdot (-4) = (16x^2)(-4), \text{ so } g'(2) = (-8)^2 \cdot (-4) = 64(-4) = -256.$$

34 Let $g(x) = \int_{-1}^{-2x} t^2 dt$. Find $g'(1)$ using FTC Part 1 and the Chain Rule.

- a. -10
- b. -8 ✓
- c. -2
- d. 4

$$g'(x) = (-2x)^2 \cdot (-2) = (4x^2)(-2), \text{ so } g'(1) = (-2)^2 \cdot (-2) = 4(-2) = -8.$$

35 Let $g(x) = \int_0^{3x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

- a. 48
- b. 435
- c. 432 ✓
- d. 144

$$g'(x) = (3x)^2 \cdot (3) = (9x^2)(3), \text{ so } g'(4) = (12)^2 \cdot (3) = 144(3) = 432.$$

36 Let $g(x) = \int_1^{-3x} t^2 dt$. Find $g'(2)$ using FTC Part 1 and the Chain Rule.

a. -12

b. 36

c. -111

d. -108 ✓

$$g'(x) = (-3x)^2 \cdot (-3) = (9x^2)(-3), \text{ so } g'(2) = (-6)^2 \cdot (-3) = 36(-3) = -108.$$

37 Let $g(x) = \int_2^{-4x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. -1028

b. 256

c. -1024 ✓

d. -64

$$g'(x) = (-4x)^2 \cdot (-4) = (16x^2)(-4), \text{ so } g'(4) = (-16)^2 \cdot (-4) = 256(-4) = -1024.$$

38 Let $g(x) = \int_{-3}^{-3x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

a. -246

b. -243 ✓

c. -27

d. 81

$$g'(x) = (-3x)^2 \cdot (-3) = (9x^2)(-3), \text{ so } g'(3) = (-9)^2 \cdot (-3) = 81(-3) = -243.$$

39 Let $g(x) = \int_3^{2x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

- a. 74
- b. 36
- c. 18
- d. 72 ✓

$$g'(x) = (2x)^2 \cdot (2) = (4x^2)(2), \text{ so } g'(3) = (6)^2 \cdot (2) = 36(2) = 72.$$

40 Let $g(x) = \int_1^{-3x} t^2 dt$. Find $g'(1)$ using FTC Part 1 and the Chain Rule.

- a. -27 ✓
- b. 9
- c. -30
- d. -3

$$g'(x) = (-3x)^2 \cdot (-3) = (9x^2)(-3), \text{ so } g'(1) = (-3)^2 \cdot (-3) = 9(-3) = -27.$$

41 Let $g(x) = \int_1^{-5x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

- a. -2000 ✓
- b. 400
- c. -80
- d. -2005

$$g'(x) = (-5x)^2 \cdot (-5) = (25x^2)(-5), \text{ so } g'(4) = (-20)^2 \cdot (-5) = 400(-5) = -2000.$$

42 Let $g(x) = \int_2^{-3x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

- a. 144
- b. -432 ✓
- c. -435
- d. -48

$$g'(x) = (-3x)^2 \cdot (-3) = (9x^2)(-3), \text{ so } g'(4) = (-12)^2 \cdot (-3) = 144(-3) = -432.$$

43 Let $g(x) = \int_0^{-5x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

a. 225

b. -1125 ✓

c. -45

d. -1130

$$g'(x) = (-5x)^2 \cdot (-5) = (25x^2)(-5), \text{ so } g'(3) = (-15)^2 \cdot (-5) = 225(-5) = -1125.$$

44 Let $g(x) = \int_{-3}^{4x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. 64

b. 1028

c. 1024 ✓

d. 256

$$g'(x) = (4x)^2 \cdot (4) = (16x^2)(4), \text{ so } g'(4) = (16)^2 \cdot (4) = 256(4) = 1024.$$

45 Let $g(x) = \int_{-2}^{2x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

a. 74

b. 36

c. 72 ✓

d. 18

$$g'(x) = (2x)^2 \cdot (2) = (4x^2)(2), \text{ so } g'(3) = (6)^2 \cdot (2) = 36(2) = 72.$$

46 Let $g(x) = \int_3^{2x} t^2 dt$. Find $g'(1)$ using FTC Part 1 and the Chain Rule.

- a. 2
- b. 10
- c. 4
- d. 8 ✓

$$g'(x) = (2x)^2 \cdot (2) = (4x^2)(2), \text{ so } g'(1) = (2)^2 \cdot (2) = 4(2) = 8.$$

Multiple choice — hard 15

47 Evaluate $\int_0^1 2x(x^2 + 2)^3 dx$ using $u = x^2 + 2$.

- a. $\frac{65}{4}$ ✓
- b. $\frac{19}{4}$
- c. $\frac{65}{3}$
- d. $-\frac{65}{4}$

$$du = 2x dx. \text{ When } x = 0, u = 2; \text{ when } x = 1, u = 3. \int_2^3 u^3 du = \left[\frac{u^4}{4} \right]_2^3 = \frac{81}{4} - \frac{16}{4} = \frac{65}{4} = \frac{65}{4}.$$

48 Evaluate $\int_0^1 2x(x^2 + 1)^2 dx$ using $u = x^2 + 1$.

- a. 1
- b. $\frac{7}{3}$ ✓
- c. $\frac{7}{2}$
- d. $-\frac{7}{3}$

$$du = 2x dx. \text{ When } x = 0, u = 1; \text{ when } x = 1, u = 2. \int_1^2 u^2 du = \left[\frac{u^3}{3} \right]_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3} = \frac{7}{3}.$$

49 Evaluate $\int_0^1 2x(x^2 + 4)^3 dx$ using $u = x^2 + 4$.

a. $-\frac{369}{4}$

b. $\frac{369}{4}$ ✓

c. 123

d. $\frac{61}{4}$

$du = 2x dx$. When $x = 0$, $u = 4$; when $x = 1$, $u = 5$. $\int_4^5 u^3 du = \left[\frac{u^4}{4}\right]_4^5 = \frac{625}{4} - \frac{256}{4} = \frac{369}{4} = \frac{369}{4}$.

50 Evaluate $\int_0^1 2x(x^2 + 4)^2 dx$ using $u = x^2 + 4$.

a. $\frac{61}{2}$

b. 3

c. $\frac{61}{3}$ ✓

d. $-\frac{61}{3}$

$du = 2x dx$. When $x = 0$, $u = 4$; when $x = 1$, $u = 5$. $\int_4^5 u^2 du = \left[\frac{u^3}{3}\right]_4^5 = \frac{125}{3} - \frac{64}{3} = \frac{61}{3} = \frac{61}{3}$.

51 Evaluate $\int_0^1 2x(x^2 + 1)^3 dx$ using $u = x^2 + 1$.

a. $-\frac{15}{4}$

b. $\frac{7}{4}$

c. 5

d. $\frac{15}{4}$ ✓

$du = 2x dx$. When $x = 0$, $u = 1$; when $x = 1$, $u = 2$. $\int_1^2 u^3 du = \left[\frac{u^4}{4}\right]_1^2 = \frac{16}{4} - \frac{1}{4} = \frac{15}{4} = \frac{15}{4}$.

52 Find the average value of $f(x) = -2x^2 - 4x + 5$ on $[0, 6]$.

a. -30

b. -91

c. -31 ✓

d. -32

Average $= \frac{1}{6} \int_0^6 (-2x^2 - 4x + 5) dx = \frac{1}{6} \left[-2\frac{x^3}{3} - 4\frac{x^2}{2} + 5x \right]_0^6 = -31$.

53 Find the average value of $f(x) = x^2 + 4x + 3$ on $[0, 6]$.

a. 27 ✓

b. 63

c. 28

d. 26

$$\text{Average} = \frac{1}{6} \int_0^6 (x^2 + 4x + 3) dx = \frac{1}{6} \left[\frac{x^3}{3} + 4\frac{x^2}{2} + 3x \right]_0^6 = 27.$$

54 Find the average value of $f(x) = x^2 - 4x$ on $[0, 2]$.

a. -4

b. $-\frac{5}{3}$

c. $-\frac{8}{3}$ ✓

d. $-\frac{11}{3}$

$$\text{Average} = \frac{1}{2} \int_0^2 (x^2 - 4x) dx = \frac{1}{2} \left[\frac{x^3}{3} - 4\frac{x^2}{2} \right]_0^2 = -\frac{8}{3}.$$

55 Find the average value of $f(x) = -3x^2 - 4x$ on $[0, 5]$.

a. -95

b. -36

c. -35 ✓

d. -34

$$\text{Average} = \frac{1}{5} \int_0^5 (-3x^2 - 4x) dx = \frac{1}{5} \left[-3\frac{x^3}{3} - 4\frac{x^2}{2} \right]_0^5 = -35.$$

56 Find the average value of $f(x) = 3x^2 + x - 3$ on $[0, 8]$.

- a. 64
- b. 66
- c. 197
- d. 65 ✓

$$\text{Average} = \frac{1}{8} \int_0^8 (3x^2 + x - 3) dx = \frac{1}{8} \left[3 \frac{x^3}{3} + \frac{x^2}{2} - 3x \right]_0^8 = 65.$$

57 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and negative for $t < -2$ and positive for $t > -2$. What feature does g have at $x = -2$?

- a. An inflection point, since g is always increasing when its integrand is positive.
- b. Neither -- accumulation functions never have local extrema.
- c. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$. ✓
- d. A local maximum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

By FTC Part 1, $g'(x) = f(x)$. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

58 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and negative for $t < 5$ and positive for $t > 5$. What feature does g have at $x = 5$?

- a. An inflection point, since g is always increasing when its integrand is positive.
- b. A local minimum, since f changes sign from negative to positive at $t = 5$, so $g' = f$ changes from $-$ to $+$. ✓
- c. A local maximum, since f changes sign from negative to positive at $t = 5$, so $g' = f$ changes from $-$ to $+$.
- d. Neither -- accumulation functions never have local extrema.

By FTC Part 1, $g'(x) = f(x)$. A local minimum, since f changes sign from negative to positive at $t = 5$, so $g' = f$ changes from $-$ to $+$.

- 59 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and positive for $t < 1$ and negative for $t > 1$. What feature does g have at $x = 1$?
- a. Neither -- accumulation functions never have local extrema.
 - b. A local minimum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.
 - c. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$. ✓
 - d. An inflection point, since g is always increasing when its integrand is positive.

By FTC Part 1, $g'(x) = f(x)$. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

- 60 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and negative for $t < -5$ and positive for $t > -5$. What feature does g have at $x = -5$?
- a. A local minimum, since f changes sign from negative to positive at $t = -5$, so $g' = f$ changes from $-$ to $+$. ✓
 - b. Neither -- accumulation functions never have local extrema.
 - c. A local maximum, since f changes sign from negative to positive at $t = -5$, so $g' = f$ changes from $-$ to $+$.
 - d. An inflection point, since g is always increasing when its integrand is positive.

By FTC Part 1, $g'(x) = f(x)$. A local minimum, since f changes sign from negative to positive at $t = -5$, so $g' = f$ changes from $-$ to $+$.

61 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and positive for $t < 5$ and negative for $t > 5$. What feature does g have at $x = 5$?

a. A local maximum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$. ✓

b. An inflection point, since g is always increasing when its integrand is positive.

c. A local minimum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.

d. Neither -- accumulation functions never have local extrema.

By FTC Part 1, $g'(x) = f(x)$. A local maximum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.